



Tomato classification with YOLOv8: Enhancing automated sorting and quality assessment

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ABSTRACT

This study presents the design and implementation of an automated system for sorting and measuring kidney tomatoes using a YOLOv8 model with a size estimation algorithm. The proposed system integrates computer vision and deep learning with a physical sorting mechanism to categorize tomatoes into three classes: green, red, and damaged, while also determining their size. The classification model was trained on a dataset of 2,145 images of tomatoes taken from different sources and lighting conditions to enhance performance during training. The implemented prototype consists of a conveyor belt equipped with sensors and a high-resolution camera to capture and analyse tomato characteristics in real-time. A servo-driven sorting mechanism then directs the classified tomatoes into their respective bins. Experimental validation and testing show that the model achieves a classification accuracy of 99.6% and a size estimation accuracy of 97.1%, aiding in a reliable and efficient post-harvest sorting process. The proposed system not only reduces the probability of human error but also improves the precision of tomato classification. Future developments will focus on refining and adapting existing AI methodologies to improve their effectiveness in agricultural environments. This includes enhancing model robustness, improving classification accuracy under real-world conditions, and tailoring AI tools to better meet the demands of industrial tomato sorting.

1. Introduction

In recent years, the integration of automation and computer vision technologies in agriculture has demonstrated significant potential in reducing costs and enhancing productivity. These technologies enable accurate identification of fruit and vegetable ripeness, optimizing harvesting time and ensuring higher-quality produce. Additionally, computer vision, useful for extracting information for images [1], can detect and diagnose plant diseases early, allowing for timely interventions that prevent crop losses while reducing labour costs and increasing operational efficiency. By enhancing precision in both harvesting and disease management, these advancements contribute to more sustainable and profitable agricultural practices.

Tomatoes are globally valued for their commercial importance but face significant challenges due to limited genetic diversity, that is, the

variety of genetic characteristics within the tomato species. Although there are many visible tomato varieties, most commercial types come from a narrow genetic pool, making them more susceptible to pests, diseases, and environmental stress despite their external differences [2]. Among tomato varieties, kidney tomatoes hold a prominent position in Ecuador's agricultural industry due to their strong demand both domestically and internationally. Economic, environmental, and social factors shape the sustainability of tomato farming in Ecuador. Farms excelling in social and economic sustainability tend to optimise conventional resources effectively, while those prioritising environmental sustainability tend to implement better ecological resource management [3,4]. The integration of automation and computer vision in Ecuadorian tomato farming could further enhance sustainability by improving efficiency, reducing waste, and promoting smarter resource utilization.

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Table 1

Classification of kidney tomatoes by size, adapted from Ecuadorian Technical Standard 1745. These criteria guide the physical sorting stage in the developed prototype.

Type [Size]	Diameter [mm]	
	Minimum	Maximum
I-Big	71	≥ 71
II-Medium	56	70
III-Small	40	55

The cultivation and distribution of this sought-after product occur predominantly within Ecuador. At the same time, tomatoes contribute to rural Ecuadorians' nutritional landscape, where local production supports better nutrient intake than market purchases [5]. However, the production and distribution of kidney tomatoes involve a complex sequence of pre- and post-harvest processes.

Pre-harvest activities encompass a range of tasks, including soil preparation, meticulous seed selection, and precise sowing or transplanting techniques. These measures are essential to ensure the quality and yield of tomatoes [6]. Regarding the post-harvest stage, which is the focus of this study, the selection and classification of the tomatoes are essential processes [7]. These are done manually, and there are different ways to classify the tomato by size, quality, and maturity. The Ecuadorian Technical Standard 1745 establishes the general requirements for the consumption or commercialisation of kidney tomatoes. Among these is the colouration that the tomato must have to be classified as green, yellow, or red. This study focuses on green and red kidney tomatoes, as defined by the Ecuadorian technical standard [8]. Green tomatoes are those whose surface area is more than 90% green, while red tomatoes have more than 90% of their surface area showing red colouration. Another criterion for classification is size; tomatoes can be graded based on their diameter, as shown in Table 1.

This study addresses the open-ended challenge of developing affordable, scalable, and robust post-harvest classification systems capable of reliable operation under diverse and unpredictable field conditions. Current manual sorting practices result in high labour costs, inconsistent quality, and significant post-harvest losses, undermining sustainable food production goals. By advancing automated tomato classification with integrated size estimation, this research directly supports the targets of Sustainable Development Goals 9 (Industry, Innovation, and Infrastructure) and 12 (Responsible Consumption and Production), promoting efficiency, reducing waste, and improving food supply chain sustainability [9].

1.1. Related works

The agricultural sector is experiencing a transformative shift due to the integration of advanced Artificial Intelligence (AI) technologies such as Convolutional Neural Networks (CNNs), Machine Learning (ML), and Computer Vision (CV). These technologies have been instrumental in addressing traditional challenges in resource optimization, crop yield improvement, and operational efficiency [10–16]. With applications ranging from crop monitoring and soil analysis to pest detection and yield prediction, AI has become a cornerstone of modern agriculture. For instance, Singh et al. [17] utilized multilayer CNNs to detect mango leaf anthracnose disease, achieving an impressive accuracy rate of 97.13%, which surpassed that of alternative methods such as Particle Swarm Optimization.

In particular, automated classification and maturity assessment of crops like tomatoes using CNNs and other AI models has inspired several studies. Khan et al. [18] created a CNN-Transformer hybrid model for classifying tomato maturity, resulting in a 94.2% accuracy with precision and recall values of 94.0%. Additionally, Begum et al. [19]

demonstrated the effectiveness of CNNs by achieving a 97.37% accuracy using the VGG-19 architecture for tomato maturity classification. Zhang et al. [20] focused on classifying tomatoes based on colour, attaining a detection accuracy of 91.9%. Lastly, Kussul et al. [21] employed deep learning for crop detection, achieving an accuracy of 85.0% with a four-level neural network architecture. These findings underscore the ability of CNNs to provide high accuracy in agricultural applications, particularly in disease detection and maturity evaluation.

Another algorithm based on CNNs architecture is the YOLO (You Only Look Once) model, which is used for real-time object detection. It processes an entire image in a single forward pass, dividing it into a grid and simultaneously predicting bounding boxes and object classes, enabling fast and accurate detection [22]. For example, Wang et al. [23] developed a lightweight YOLOv5n-based model for real-time cherry tomato detection, achieving 95.0% accuracy, 92.0% recall, and 95.2% mean average precision (mAP) at a detection speed of 5.3 ms. In a similar study, Zheng et al. [24] introduced an improved YOLOv8-Tomato model for recognizing tomatoes under greenhouse conditions, achieving 99.4% mAP and 99.0% recall. Also, Yang et al. [25] enhanced YOLOv8s with modules such as Depthwise Separable Convolution and Dual-Path Attention Gate, achieving a mAP of 93.4% and a real-time detection speed of 138.8 FPS. Vo et al. [26] leveraged YOLOv9 for automated tomato ripeness classification, achieving a precision of 85.6% and a recall of 83.2%, demonstrating the potential of integrating robotics with AI for automated agricultural monitoring. Similarly, Do et al. [27] applied YOLOv5s to automate tomato classification, achieving an accuracy of approximately 90.2% on the Tomato dataset, underscoring the robustness of AI-driven solutions for real-world agricultural applications. These advancements highlight YOLO models' growing dominance in agricultural AI.

The integration of AI models with robotic systems, as Unmanned Aerial Vehicles [28], has further revolutionized tomato harvesting and sorting. Angeles et al. [29] combined YOLOv5 and YOLOv8 models with robotic manipulators for post-harvest tomato sorting, achieving a detection accuracy of 99.5% and classification precision of 99.1%. Rajesh et al. [30] employed YOLOv8 for greenhouse tomato harvesting, achieving 97.23% accuracy in distinguishing ripe and unripe tomatoes, even under challenging conditions. In addition, for a more in-depth contribution, Li et al. [31] developed the MHSA-YOLOv8 model for grading and counting tomatoes, achieving an mAP of 86.4% for grading and 99% precision for counting, demonstrating its efficiency in unmanned agricultural operations. These studies illustrate the synergy between AI and robotics in advancing automation in agriculture.

In addition to this kind of models, other machine learning approaches have been explored. For example, Phan et al. [32] investigated tomato classification into ripe, immature, and damaged categories using YOLOv5m, ResNet50, ResNet-101, and EfficientNet-B0. Trained on 4500 images with 200 epochs, the models demonstrated high performance, with YOLOv5m and ResNet-101 achieving 100.0% accuracy for ripe and immature tomatoes and EfficientNet-B0 reaching 94.0% for damaged fruit. Also, Behera et al. [33] explored automated tomato classification and grading based on size, shape, and colour using image processing techniques. Their MATLAB-based system achieved 96.67% classification accuracy and 100% grading precision, showcasing the potential of image-based AI solutions.

Advancements in lightweight AI models have expanded their applicability for real-time agricultural monitoring. Zheng et al. [34] enhanced YOLOv5 by incorporating MobileNetV3's bottleneck module and genetic algorithm optimization, achieving a mean Average Precision (mAP) of 96.9% with a 42.5 ms detection speed on a CPU. Their quantized model reached 26.5 FPS on mobile devices while maintaining 93.0% accuracy, significantly reducing computational demands. Liu et al. [35] improved YOLOv3 with a dense architecture and circular bounding boxes for tomato detection, demonstrating enhanced robustness under varying lighting and occlusion conditions. Camacho et al. [36] compared Mask R-CNN and YOLOv8 for tomato maturity segmen-

tation, with YOLOv8 achieving superior performance across ripeness categories.

Beyond CNNs and YOLO, numerous studies highlight the significance of AI in tomato classification. Elbendary et al. [37] employed Principal Component Analysis (PCA) and Support Vector Machines (SVMs) for tomato ripeness classification, achieving a 90.80% accuracy with a one-against-one SVM approach. Pacheco et al. [38] evaluated K-Nearest Neighbors (K-NN), Multi-Layer Perceptron (MLP), and K-Means Clustering for tomato maturity classification, reporting over 90.0% accuracy for supervised methods. Kaur and Gupta [39] utilized Random Forest for maturity classification, attaining 92.49% accuracy and outperforming SVM, which achieved only 70.88%. Mputu et al. [40] introduced a hybrid model integrating CNN-based feature extraction with SVM classifiers, achieving 97.50% accuracy in binary classification and 96.67% in multi-class tasks, underscoring the robustness of hybrid approaches.

Overall, the integration of AI, particularly CNNs and YOLO architectures, has substantially advanced tomato classification, maturity assessment, and automation in agriculture. These innovations not only enhance precision and efficiency but also pave the way for scalable, real-time solutions applicable to various agricultural domains.

1.2. Main contributions

This work focuses on automating post-harvest classification and sorting processes for kidney tomatoes within an industrial assembly line. In Ecuador, manual sorting remains the prevailing practice, despite being labour-intensive, time-consuming, and prone to human error, which often results in inefficiencies and economic losses. To address these challenges, the authors propose an integrated, real-time sorting system that utilises a YOLOv8-based deep learning model trained to classify ripeness levels and detect surface damage, along with a computer vision module for size estimation. This dual-parameter approach improves grading precision and aligns with the operational demands of high-throughput tomato processing environments. By minimising human intervention and enhancing classification speed and accuracy, the proposed system directly responds to critical limitations observed in current post-harvest handling practices.

In response to these limitations, the present study introduces a fully integrated hardware–software framework, delivering several key contributions to the field of automated agricultural processing. These contributions are summarised as follows:

- We propose a novel low-cost architecture that integrates computer vision and mechanical actuation for post-harvest tomato classification, demonstrating its viability in semi-industrial settings with consistent sorting performance.
- We contribute a dual-parameter classification framework that combines ripeness detection (green, red, damaged) with size estimation, providing a more realistic approach aligned with market quality standards.
- We demonstrate that a YOLOv8-based model, trained on a domain-diverse dataset, achieves 99.6% classification accuracy and 97.1% size estimation accuracy under real-time operating constraints, outperforming comparable state-of-the-art solutions.
- We deliver experimental insights on the impact of lighting variability and surface defects on tomato classification, highlighting critical challenges and directions for achieving robust deployment in real-world agricultural environments.

The structure of this paper is divided as follows: Section 1 describes the introduction, the state of the art, and the main contribution of this work. Next, Section 2 presents the materials and methods that contain a detailed description of the prototype design that includes the general architecture, and the hardware and software components. Then, Section 3 describes the experiments and results of this work. This section consists of the calibration, validation and testing steps of the model. In Section 4,

the performance of the machine and software is described. Section 5 details the discussion of this work, and Section 6 lists the main conclusions of the present research.

2. Materials and methods

This section outlines the proposed architecture and the hardware and software components of the developed prototype, which enables the sorting of tomatoes based on ripeness and size. First, a general description of the architecture is presented. Then, a detailed explanation of the hardware components used in constructing the machine will be provided. Finally, the software components and the algorithms used for ripeness classification and size sorting are presented and explained.

2.1. General architecture

The proposed architecture for the tomato sorting machine incorporates a conveyor belt mechanism driven by a stepper motor, along with sensors and a high-resolution camera. The camera captures real-time images of the tomatoes, which are then processed by a computer vision-based system. The fully developed prototype is depicted in Fig. 1.

The tomato classification algorithm combines the YOLOv8 algorithm with a pixel-based computer vision system to detect the fruit's ripeness grade and then sort it according to its size. Once classification is complete, the system uses servomotors to guide each tomato to its designated sorting bin based on the results. This process eliminates the need for manual intervention, minimises human error, and improves the efficiency and accuracy of post-harvest sorting operations. The architecture's hardware and software integration ensure a streamlined and scalable solution suitable for industrial applications. In particular, the system was designed to operate as part of a scalable assembly line, enabling continuous tomato flow and automatic decision-making during sorting. By capturing and processing each tomato in real-time and executing classification and redirection within a single, uninterrupted sequence, the machine performs quality assessment and physical sorting seamlessly. This design aligns with post-harvest industrial requirements, ensuring that the system can be integrated into small or high-throughput environments with minimal adaptation.

2.2. Hardware components

The implemented prototype incorporates several key hardware components to ensure the correct operation and execution of the sorting task. Each element has been selected and integrated to perform specific tasks within the classification process. Fig. 2 illustrates the main components of the machine, which are described in detail below.

A conveyor belt carries the tomatoes through the sorting system, powered by a stepper motor. This design allows for consistent spacing and orientation of tomatoes as they pass through the classification zone. In addition, sensors are strategically placed along the conveyor path to detect the presence of tomatoes and ensure proper positioning for image capture. These sensors trigger the camera and provide real-time feedback to the control system, enabling synchronised operation between hardware and software components. Moreover, a high-resolution camera is installed above the conveyor belt to capture images of each tomato as it moves through the system. The camera's placement ensures optimal image quality, with a clear view of the tomato's surface for computer vision analysis. Controlled lighting conditions are maintained to minimise shadows and reflections, enhancing the accuracy of image-based classification.

Once the software classifies a tomato based on ripeness and size, the sorting mechanism directs it to an indicated storage unit. This mechanism consists of servomotors that move physical barriers or flaps to guide the tomato into one of the designated collection areas. The servomotors respond to commands from the control system, executing movements to ensure accurate sorting without damaging the tomatoes.

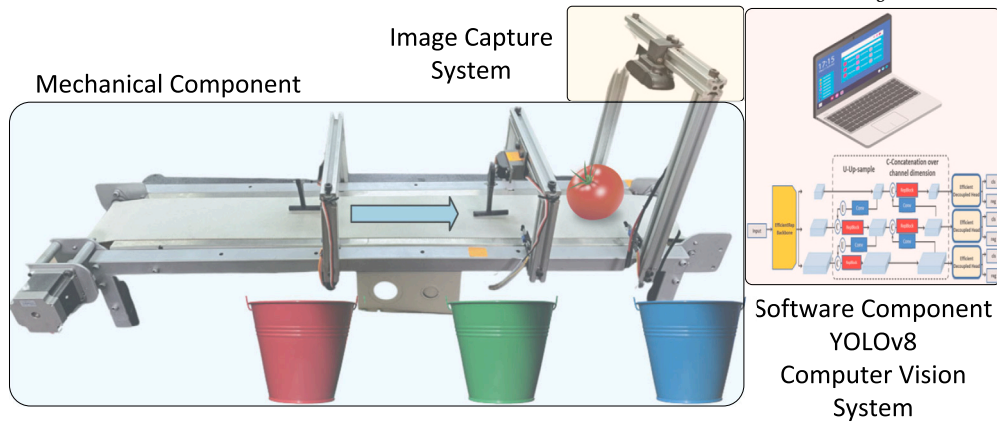


Fig. 1. Prototype for tomato sorting. The machine includes a conveyor belt, camera, proximity sensors, and servo-controlled gates for automated classification based on colour and size.

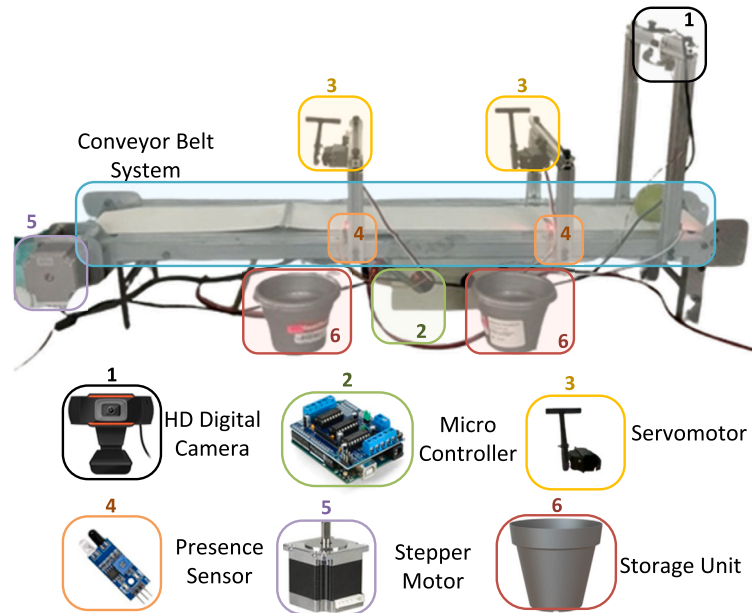


Fig. 2. Hardware components of the tomato sorting system. The image highlights the main physical elements, including the stepper motor, high-resolution camera, sensors, and sorting gates.

2.3. Software components

This subsection details the software framework developed for the tomato sorting machine, which integrates computer vision and machine learning techniques. The YOLOv8 model is responsible for identifying tomato ripeness and bounding box. The model development includes dataset acquisition, labelling, preparation, and training and validation of the YOLOv8 algorithm. Additionally, a size determination algorithm was implemented (as shown in Section 2.4.1), leveraging the bounding box data provided by the model to categorise tomatoes based on their dimensions. Fig. 3 presents a diagram of the software component for this work.

2.3.1. Data collection and dataset creation

The dataset for this study was created from 2,145 images, gathered from various sources to ensure diversity. The images were captured under varying lighting conditions to cover three primary classes: ripe, unripe, and damaged tomatoes.

Each image was manually annotated using Roboflow, which allowed for efficient tagging of bounding boxes around each tomato according to its class. The dataset was divided into three subsets: 70% for training,

Table 2

Distribution of the tomato image dataset used for model training, validation, and testing.

Category	Images
Training (70%)	1500
Validation (20%)	429
Testing (10%)	216

20% for validation, and 10% for testing. This split ensured an ample amount of data for training while retaining sufficient unseen examples for evaluation and validation, as shown in Table 2.

2.3.2. YOLOv8 model for classification by maturity level

The YOLOv8 model was selected for the task of tomato maturity level classification due to its advanced real-time object detection capabilities, which balance high accuracy with computational efficiency [22]. Its design and architecture enable effective operation in dynamic and resource-constrained environments, making it ideal for agricultural applications.

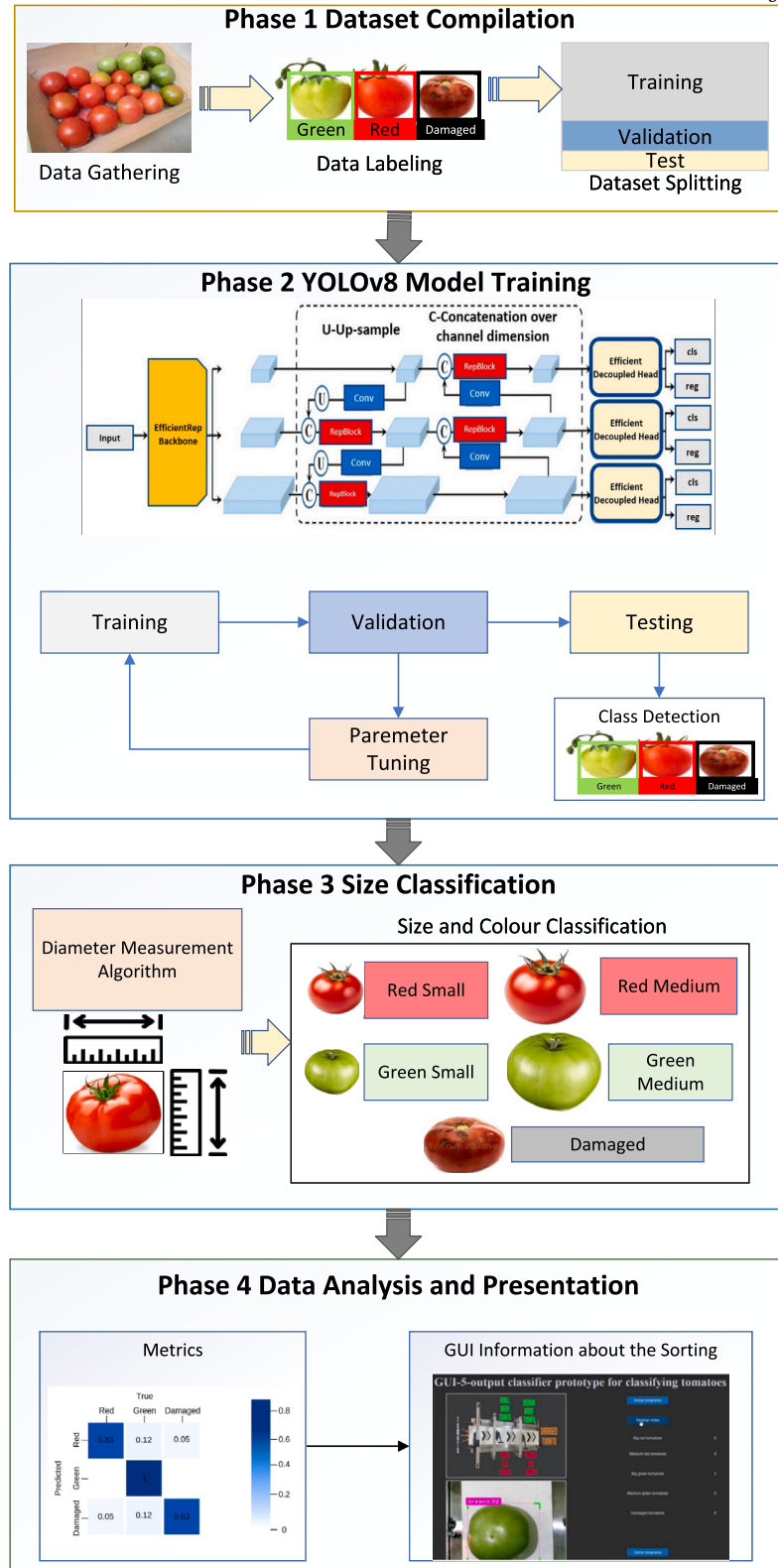


Fig. 3. Flow diagram of the intelligent tomato sorting system. The diagram shows the integration of YOLOv8-based detection and size estimation into the sorting decision logic.

YOLOv8 introduces significant improvements over its predecessors, such as YOLOv5 and YOLOv7. The architecture of YOLOv8 features two primary components: the backbone and the head. The backbone is based on a modified CSPDarknet53 framework, which integrates Cross-Stage Partial connections for improved feature reuse and reduced

computational overhead. These enhancements support detailed feature extraction, crucial for identifying subtle variations in tomato ripeness [41]. The head section of the YOLOv8 model leverages a self-attention mechanism that dynamically adjusts feature importance, ensuring robust detection under varying environmental conditions. By adopting

Table 3

Parameters explored during YOLOv8 training. Various model sizes, optimizers (SGD, Adam, AdamW), epochs, and batch sizes were tested to identify the optimal configuration for tomato classification and size estimation.

Parameter	Value
Model Size	YOLOv8 Nano - 3.15M parameters
	YOLOv8 Small - 11.2M parameters
Epochs	25
	50
	75
	100
Optimizer	Stochastic Gradient Descent (SGD)
	Adaptive Moment Estimation (Adam)
	Adaptive Moment Estimation with Weight Decay (AdamW)
	8
Batch Size	16
	32
	64
Starting Weights	YOLOv8 Default
Number of Classes	3
Learning Rate	Automatically adjusted by model

an anchor-free detection approach, YOLOv8 simplifies the network architecture and enhances its ability to detect small objects, such as individual tomatoes in cluttered settings. This design choice minimizes false positives and accelerates non-maximum suppression during post-processing, aligning with modern efficiency demands [22]. YOLOv8's advancements are underpinned by its adoption of the C2f module, which facilitates contextual information integration and improves feature fusion across multiple scales. These features collectively position YOLOv8 as a state-of-the-art solution for tasks requiring precise classification, such as maturity-level detection in agricultural produce [42].

2.4. Training

A testing environment was implemented once the datasets were defined and correctly labelled. The training environment was transitioned to Google Colab with 12,7GB de RAM and T4 GPU with 15GB VRAM. In pursuit of optimal accuracy in the detection and classification of green, red and damaged tomatoes, a variety of training variations were implemented. The specific parameters utilized during the training phase are outlined in Table 3. The models used in the training were YOLOv8 Nano and YOLOv8 Small. The Nano version has 3.15M parameters, achieving mAP@50 of 42% and mAP@75 of 34%. It offers the fastest inference speed and is ideal for mobile, edge, and fast prototyping scenarios with minimal RAM/VRAM, favouring speed. In contrast, the small model (11.2M parameters) delivers higher accuracy (mAP@50 56%, mAP@75 48%) and is better suited for edge applications needing improved detection, balancing speed with moderate resource demands.

Once attempting various configurations, YOLOv8 small with SGD optimizer and batch size of 16 was employed as the most effective in this application. Though larger versions of the model provided marginally better precision, the improvement came at the cost of considerably longer training times and higher computational demand. On average, the faster models, the smaller ones, did particularly poorly in identifying rotten tomatoes, which are more difficult due to their less regular features. In terms of the size of the batches, higher values produced slower convergence but extremely small batches created instability during training. More stable behaviour over epochs was provided by SGD compared to adaptive optimizers like Adam or AdamW. Throughout the training stage, the model was monitored for performance with validation loss, which is the YOLOv8 default. Automatic early stopping was not employed, but training was manually terminated when the validation loss was observed to show no improvement beyond a certain point after approximately 100 epochs.

2.4.1. Size detection algorithm

Following maturity classification, a size detection algorithm was developed to sort tomatoes into predefined size categories, considering the fixed camera height used for all image captures. The development of this algorithm involved the following steps:

1. Bounding Box Extraction: YOLOv8 outputs bounding box coordinates for each detected object. These coordinates (X_1 , Y_1 , X_2 , Y_2) were extracted for each detected tomato. The dimensions of the bounding box, representing the width and height of the detected object in pixels, were calculated as:

$$\begin{aligned} \text{Width} &= X_2 - X_1, \\ \text{Height} &= Y_2 - Y_1. \end{aligned} \quad (1)$$

2. Real-World Dimension Mapping: A calibration process was conducted by placing the prototype's digital camera at a fixed distance of 21.5 cm. A picture of a reference object with known dimensions was taken at the specified distance. Then, a scaling factor was obtained to convert pixel measurements into millimetres by correlating the pixels that the object measures in the pictures with its real-world dimensions.
3. Size Classification: Using the calibrated dimensions, the tomatoes were categorized into two size classes: small and medium. The thresholds for these categories were determined based on the Ecuadorian technical standard for tomato classification [8], which specifies diameter ranges for each size class. It is important to note that no size detection is performed if the tomato is detected as damaged.
4. Testing and Adjustment: The size classification algorithm was tested extensively using tomatoes of known dimensions. Based on this evaluation, the complete procedure was structured to ensure reliable estimation of tomato size and sorting. The process integrates maturity assessment with image analysis. The YOLOv8 model detects tomatoes and provides bounding box coordinates, from which the width and height in pixels are calculated. These dimensions are converted to centimetres using scaling factors of 0.023 cm/pixel (width) and 0.021 cm/pixel (height), values adjusted experimentally to match manual calliper measurements closely. A confidence threshold of 0.85 ensures reliable detections. Finally, each tomato is assigned to a size category: those with an estimated diameter below 56 mm are classified as small. In comparison, those equal to or greater than 56 mm are classified as medium. The whole sequence, including the conversion formulas and sorting logic, is summarised in Algorithm 1. Results were compared to manual measurements with a digital calliper to validate accuracy, as shown in the Fig. 4. Minor adjustments were made to the thresholds to improve consistency with real-world observations.

2.5. Resource optimization

The automated tomato sorting system is specifically designed to optimise key agricultural resources by reducing reliance on manual labour, minimising waste, and increasing throughput efficiency. First, the integration of YOLOv8 enables rapid and reliable classification of tomato ripeness and damage conditions directly on the assembly line, eliminating the need for manual visual inspection. This substantially reduces labour costs and human error, particularly during peak harvest seasons when workforce availability may be limited.

Second, the system's inference-time operation ensures that tomatoes are sorted with minimal delay, with a total processing time of approximately 5-8 seconds per fruit, thereby reducing exposure to conditions that could cause spoilage or bruising. The size estimation algorithm prevents improper bin allocation, which often leads to product downgrading or rejection. By aligning classification thresholds with Ecuadorian

Algorithm 1 Tomato Maturity Classification and Size Sorting Algorithm.

Input: Image captured at fixed camera height
 Apply YOLOv8 model to detect tomatoes and obtain:
 - Bounding box coordinates: (x_1, y_1, x_2, y_2)
 - Predicted class: $c = \arg \max P(c|x)$
 - Confidence: $P(c|x)$
for each detected tomato **do**
 Compute bounding box dimensions in pixels:
 $W_p = x_2 - x_1, H_p = y_2 - y_1$
 Convert dimensions to centimetres:
 $W_{cm} = W_p \cdot S_x, H_{cm} = H_p \cdot S_y$
 where $S_x = 0.023 \text{ cm/pixel}, S_y = 0.021 \text{ cm/pixel}$
if $P(c|x) \geq 0.85$ **then**
 if $c = \text{Green or } c = \text{Red}$ **then**
 if $W_{cm} \geq 56$ **then**
 Classify as Medium tomato
 else
 Classify as Small tomato
 end if
 else if $c = \text{Damaged}$ **then**
 Classify as Damaged tomato
 end if
 Send sorting command via serial interface
end if
end for
Output: Tomato class, size category, and sorting action



Fig. 4. Manual measurement of tomato diameter with a digital calliper. These measurements were used for calibration and validation of the size detection algorithm.

Technical Standard 1745 [8], the system ensures consistent grading and improved price recovery in the market.

Lastly, hardware resource efficiency is achieved through the use of lightweight, cost-effective components, such as stepper motor-driven conveyors, high-efficiency servomotors, and microcontroller-based logic control. The entire platform is scalable and low-power, making it suitable for deployment in both small and large processing facilities without requiring extensive infrastructure upgrades.

3. Experiments and results

This section details the findings from various configurations of the trained models. To ensure a balanced evaluation, we rely on metrics such as precision, recall, accuracy, and the F1-score. These metrics are defined mathematically, allowing their precise quantification. The corresponding formulations are presented in Equations (2) to (5), as outlined by [43]. In addition, the authors noted that they consider the exact characteristics of the expressions, specifically defining TP, TN, FP, and FN as true positives, true negatives, false positives, and false negatives, respectively.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$F1 - \text{Score} = \frac{2(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \quad (5)$$

The primary objective of this study is to find the right characteristics of a machine learning model capable of performing two specific activities: identifying the presence of a tomato fruit within a given category and determining its classification into one of the designated state-damaged, green, or red.

In the detection task, we evaluate the model's performance using four metrics, as this stage proved to be the most challenging. Leveraging a controlled machine environment, the detection model enabled precise estimation of the tomatoes' approximate dimensions. This capability was crucial for accurately identifying fruits, even under challenging conditions such as partial exposure to ambient lighting and the presence of shadows. The algorithm's robustness was further demonstrated by its ability to effectively handle the inherent complexities of the dataset. Meanwhile, in the classification section the models achieved accuracy values ranging from 97% to 100%, with the latter being the most prominent.

3.1. Validation outcomes for ripeness

For the validation phase, a dataset comprising 429 images was utilized. Consequently, adjustments were introduced to the training metrics of the machine learning model. The outcomes of these modifications are illustrated in Fig. 5, highlighting the most effective model. By systematically varying the parameters detailed in Fig. 5, this study identified an optimized model for tomato detection and classification, significantly reducing errors during the validation phase, as will be demonstrated in Section 3. The chosen model is based on the YOLOv8 small architecture, trained without dataset augmentation, utilizing 100 epochs and a batch size of 16. Additionally, the training process employed the SGD optimizer, ensuring stability and efficient convergence.

The model was also evaluated using the testing subset, which includes images never before seen by the model during training nor validation. The system performed satisfactorily, successfully detecting tomato fruits and accurately classifying them into three categories: red, green, and damaged. These findings reinforce the model's reliability and adaptability, demonstrating its effectiveness when applied to diverse image datasets.

3.2. Detection outcomes

The visualizations shown in Fig. 6 provide a detailed look at the class distribution and the spatial patterns of the detected bounding boxes, offering meaningful insights into how the object detection model performs. In 6 a), the bar chart provides an overview of how instances are distributed across the three classes: "Damaged," "Green," and "Red", showing a fairly balanced dataset. Moving to 6 b), the bounding box overlay reveals that most annotations are concentrated towards the centre of the images, which could indicate a slight bias in object positioning. Then, in 6 c), the heatmap of bounding box centres highlights where objects tend to appear most often, with a clear clustering effect near the middle. Here, the X and Y axes represent the normalized position of the bounding box centre, making it easier to compare different image sizes. Finally, 6 d) explores how bounding box sizes vary, showing common aspect ratios and trends in object dimensions. Both the width (X axis) and height (Y axis) are normalized between 0 and 1, ensuring consistency across images regardless of their original resolution. Moreover, the class distribution bar chart reveals that the dataset is well-balanced, with the damaged, green, and red classes each comprising roughly 500

Default Parameters from Ultralytics														
Model	Class	Detection Matrix				Precision	Mean Precision	Recall	Mean Recall	Accuracy	Mean Accuracy	F1 Score	Mean F1 Score	Classification Accuracy
YOLOv8 nano	Damaged	0.94	0.03	0.1	0.42	77.15%	83.85%	94.40%	93.47%	89.31%	91.88%	84.91%	88.31%	97.00%
	Green	0.0	0.96	0	0.32	88.96%		95.80%		94.58%		92.26%		100.00%
	Red	0.06	0	0.90	0.26	85.43%		90.21%		91.76%		87.76%		97.00%
	Background	0	0.01	0	0	-		-		-		-		-
YOLOv8 small	Damaged	0.89	0.01	0.03	0.4	83.00%	85.06%	88.81%	94.64%	90.63%	92.79%	85.81%	89.49%	99.00%
	Green	0.04	0.99	0	0.4	84.83%		98.60%		93.55%		90.97%		99.00%
	Red	0.07	0	0.97	0.2	87.34%		96.50%		94.20%		91.70%		97.00%
	Background	0	0.01	0	0	-		-		-		-		-
Optimal Outcomes: YOLOv8 small														
Epochs	Class	Detection Matrix				Precision	Mean Precision	Recall	Mean Recall	Accuracy	Mean Accuracy	F1 Score	Mean F1 Score	Classification Accuracy
25	Damaged	0.96	0	0.06	0.2	91.33%	91.39%	95.80%	96.50%	95.61%	95.10%	93.52%	93.87%	98.00%
	Green	0.01	0.99	0	0.48	91.61%		99.30%		96.73%		95.30%		100.00%
	Red	0.03	0	0.94	0.32	91.22%		94.41%		92.95%		92.78%		98.00%
	Background	0	0.01	0	0	-		-		-		-		-
50	Damaged	0.97	0	0.01	0.22	95.86%	94.65%	97.20%	98.37%	97.69%	97.61%	96.53%	96.46%	99.00%
	Green	0	0.99	0	0.61	92.81%		99.30%		97.24%		95.95%		100.00%
	Red	0.03	0	0.99	0.17	95.27%		98.60%		97.91%		96.91%		98.00%
	Background	0	0.01	0	0	-		-		-		-		-
75	Damaged	0.97	0	0.02	0.27	95.86%	95.86%	97.20%	97.20%	97.66%	97.66%	96.53%	96.53%	99.00%
	Green	0	0.97	0	0.55	95.83%		96.50%		97.43%		96.17%		100.00%
	Red	0.03	0	0.98	0.18	95.89%		97.90%		97.89%		96.89%		99.00%
	Background	0	0.03	0	0	-		-		-		-		-
100	Damaged	0.99	0	0.03	0.27	94.63%	96.33%	98.60%	97.20%	97.66%	97.81%	96.58%	96.75%	99.00%
	Green	0	0.97	0	0.45	96.50%		96.50%		97.66%		96.50%		100.00%
	Red	0.01	0	0.97	0.27	97.87%		96.50%		98.12%		97.18%		100.00%
	Background	0	0.03	0	0	-		-		-		-		-
Optimal Outcomes: 100 epochs														
Optimizer	Class	Detection Matrix				Precision	Mean Precision	Recall	Mean Recall	Accuracy	Mean Accuracy	F1 Score	Mean F1 Score	Classification Accuracy
SGD	Damaged	0.99	0	0.01	0.22	97.26%	97.22%	99.30%	97.90%	98.82%	98.36%	98.27%	97.56%	100.00%
	Green	0	0.96	0	0.44	97.16%		95.80%		97.67%		96.48%		100.00%
	Red	0.01	0	0.99	0.33	97.24%		98.60%		98.59%		97.92%		100.00%
	Background	0	0.04	0	0	-		-		-		-		-
ADAM	Damaged	0.94	0.01	0.03	0.41	88.74%	90.76%	93.71%	96.04%	94.06%	95.45%	91.16%	93.32%	98.00%
	Green	0.01	0.97	0	0.48	90.26%		97.20%		95.59%		93.60%		100.00%
	Red	0.05	0	0.97	0.11	93.29%		97.20%		96.71%		95.21%		98.00%
	Background	0	0.01	0	0	-		-		-		-		-
ADAMW	Damaged	0.97	0	0.03	0.32	92.62%	93.07%	96.50%	96.97%	96.30%	96.60%	94.52%	94.98%	98.00%
	Green	0.01	0.97	0	0.41	92.67%		97.20%		96.52%		94.88%		100.00%
	Red	0.02	0	0.97	0.27	93.92%		97.20%		96.97%		95.53%		100.00%
	Background	0	0.03	0	0	-		-		-		-		-
Optimal Outcomes: SGD Optimizer														
Batch Size	Class	Detection Matrix				Precision	Mean Precision	Recall	Mean Recall	Accuracy	Mean Accuracy	F1 Score	Mean F1 Score	Classification Accuracy
8	Damaged	0.99	0	0.01	0.44	95.92%	96.98%	98.60%	96.74%	98.11%	97.88%	97.24%	96.59%	100.00%
	Green	0	0.93	0	0.33	97.79%		93.01%		96.96%		95.34%		99.00%
	Red	0.01	0	0.99	0.22	97.24%		98.60%		98.57%		97.20%		100.00%
	Background	0	0.07	0	0	-		-		-		-		-
16	Damaged	0.99	0	0.01	0.43	97.24%	97.68%	98.60%	98.37%	98.60%	98.68%	97.92%	98.02%	100.00%
	Green	0	0.97	0	0.57	97.20%		97.20%		98.14%		97.20%		100.00%
	Red	0.01	0	0.99	0	98.61%		99.30%		99.29%		98.95%		100.00%
	Background	0	0.03	0	0	-		-		-		-		-
32	Damaged	0.98	0	0.01	0.29	97.22%	97.23%	97.90%	98.13%	98.36%	98.44%	97.56%	97.68%	100.00%
	Green	0	0.98	0	0.71	96.55%		97.90%		98.14%		97.22%		100.00%
	Red	0.02	0	0.99	0	97.92%		98.60%		98.83%		98.26%		98.00%
	Background	0	0.02	0	0	-		-		-		-		-
64	Damaged	0.99	0	0.01	0.11	97.92%	97.46%	98.60%	98.13%	98.83%	98.52%	98.26%	97.80%	99.00%
	Green	0	0.97	0	0.67	95.86%		97.20%		97.68%		96.53%		100.00%
	Red	0.01	0	0.99	0.22	98.60%		98.60%		99.06%		98.60%		98.00%
	Background	0	0.03	0	0	-		-		-		-		-
Optimal Outcomes: 16 batch														
Best Model Features: YOLOv8 small, 100 epochs, SGD Optimizer, 16 Batch Size														

Fig. 5. Validation results of the YOLOv8-based tomato classification model under different training configurations. The Figure summarizes precision, recall, accuracy, and F1-score across multiple experiments, varying parameters such as optimizer type (SGD, Adam, AdamW), number of epochs, model size, and batch size. The best model configuration—YOLOv8-Small, 100 epochs, batch size of 16, and SGD optimizer—achieved 99.6% classification accuracy. These results guided the selection of the final model used for deployment.

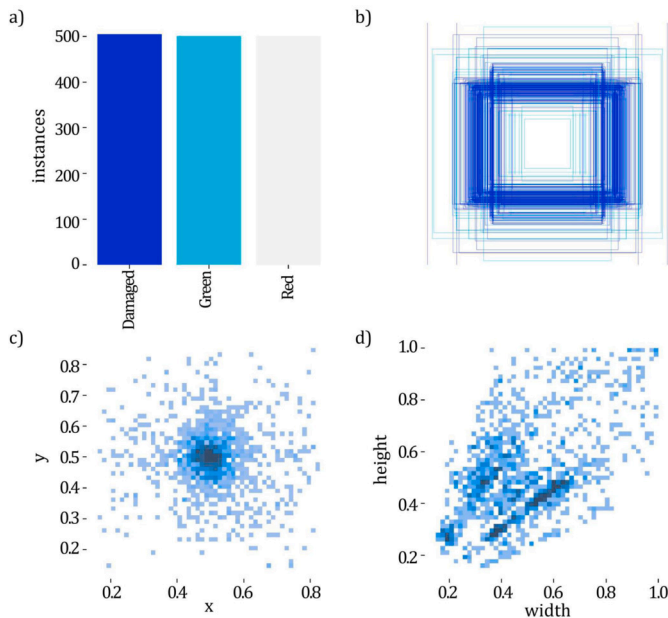


Fig. 6. Detection analysis results. (a) Class distribution bar chart; (b) spatial distribution of bounding boxes; (c) heatmap of bounding box centres; (d) bounding box width-height distribution. These plots help visualize dataset balance, object positioning, and scale variability.

instances. This balance is crucial, as it ensures no single class dominates the training process, ultimately supporting the model's ability to generalize effectively across different scenarios.

Looking more closely at the spatial distribution in Fig. 6 c) y d), the bounding box visualization highlights those objects are primarily concentrated near the centre of the images. This central clustering, evidenced by overlapping bounding boxes, suggests a bias in how objects are positioned within the dataset. Additionally, density plots for the X and Y coordinates confirm this tendency, showing a clear accumulation of bounding boxes around the middle of the image. The width-height scatter plot further complements this analysis by revealing patterns in the size of the bounding boxes. Most boxes fall within a moderate range of width and height, with smaller boxes appearing less frequently. This variability in box dimensions indicates that the model is trained on objects of different scales, which is essential for testing its ability to handle a wide range of object sizes. Such diversity is a key factor in ensuring the model's robustness and versatility in real-world applications.

3.3. Testing outcomes

After the determination of the most suitable model for the tasks outlined in this paper, its implementation was carried out and the results were subsequently presented in Fig. 7. As demonstrated, the model successfully identifies and encapsulates the tomato fruit within a bounding rectangle. Moreover, the model not only excels in object detection but also in classification, accurately categorising each object into its respective class—whether a green tomato, a red tomato, or one exhibiting signs of damage.

The model demonstrates robust performance in detecting and classifying tomatoes into green, red, and damaged categories. However, it faces certain challenges. A key difficulty was the variability in lighting conditions, which occasionally led to varying in measurements for size classification, particularly in areas affected by low brightness or shadows. Furthermore, natural variations in tomato size, shape, and surface texture introduced inconsistencies, making it more challenging to achieve precise classification concerning size. Additionally, inconsistencies within the dataset images made it difficult to accurately determine the exact boundaries of each tomato, leading to more variations in the

perceived size of the fruits. Furthermore, factors like varying tomato positions, image distortions, and the complex features of damaged fruits contribute to the limitations observed. Despite these challenges, the model achieves an overall precision of 98.61% for green tomatoes, 85.0% for red tomatoes, and 86.09% for damaged tomatoes regarding testing dataset. This set includes 216 images and the confusion matrix obtained by applying the selected model is shown in Fig. 8.

To derive the metrics from the confusion matrix, Equations (2) to (5) have been applied. Therefore, the validation and testing results of the selected model are presented in Table 4, indicating the values of key metrics commonly employed to evaluate model performance. From the results, it is evident that the model demonstrates both consistency and robust performance when assessed against the test set. Specifically, the model achieves an average of 94% across recall, precision, and F1-score, underscoring its ability to identify relevant instances while maintaining a balanced trade-off between sensitivity and specificity. The overall accuracy reaches 95.34%, further supporting the model's reliability. For the classification task, a performance of 99.6% is attained, demonstrating the model's effectiveness in distinguishing between categories. These results indicate not only high precision but also strong generalization capabilities, making the model well-suited for real-world deployment.

On the other hand, Fig. 9 presents the graphs depicting various performance metrics achieved by applying the model to the test dataset. The graphs highlight the model's impressive performance on the test dataset, showcasing high precision, recall, and F1-scores across all classes. The Precision-Recall and F1-Confidence curves reflect its balanced predictions, while the Precision-Confidence curve underscores its accuracy at lower confidence levels. As expected, recall decreases slightly as the confidence threshold rises. Overall, these results confirm the model's reliability and suitability for classification tasks.

4. Machine and software performance

The machine's structure and assembly were carefully developed in-house to ensure functionality and reliability. As shown in Fig. 10, it features a conveyor belt designed to sort tomatoes based on colour and size. Key components include a stepper motor driving the conveyor belt, a camera for tomato analysis, and servo-controlled gates that guide the fruits to their designated sections. Its lightweight aluminium frame ensures both stability and ease of handling, while a control unit below the structure allows for system adjustments and monitoring.

To operate, the machine is powered on, and tomatoes are placed on the belt. With the help of the camera and the software their condition is evaluated, and the sorting mechanism guides them to the appropriate collection area. If the tomatoes are green, they stop at the second proximity sensor and are classified according to their size. On the other hand, those in red advance to the third sensor where they are classified in the same way. Finally, the damaged advance to the end of the belt.

Regarding the software, the system includes an intuitive interface that allows users to manually initiate the classification process and monitor a real-time video feed of the detection and sorting operations. This feature enhances both transparency and operational precision. Once a tomato is detected, the software automatically calculates and plots its dimensions along the X and Y-axes, enabling accurate estimation of its physical size. Additionally, integrated digital counters register the number of tomatoes processed, categorising them by condition (ripeness or damage) and size in accordance with predefined standards.

In addition to its classification capabilities, the system was designed with a strong emphasis on resource optimisation. The hardware architecture utilises low-cost, commercially available components to ensure reliable performance while minimising energy consumption. On the software side, the use of the YOLOv8 small model provides high detection accuracy with minimal computational overhead, thereby eliminating the need for high-performance or costly processing units. Moreover, the automation of the sorting workflow reduces labour requirements and operational time, minimising human error and post-harvest prod-

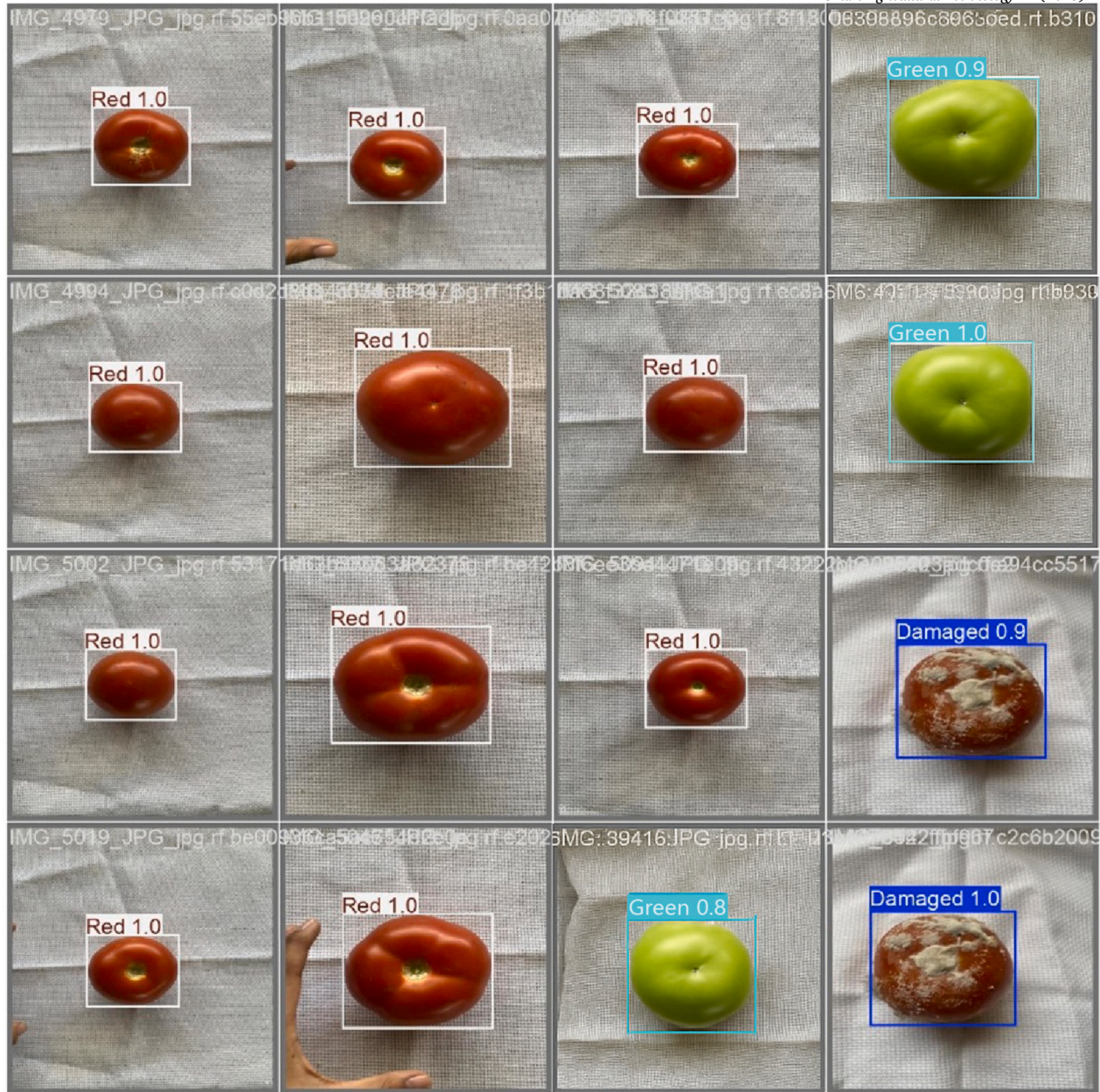


Fig. 7. Detection and classification examples. The figure shows bounding boxes with predicted labels (Damaged, Green, Red) as identified by the YOLOv8 model in real-time.

Table 4

Summary of performance metrics (precision, recall, accuracy, and F1-score) for each class during validation and testing.

Metrics	Class	Validation			Testing		
		Detection	Mean Detection	Classification	Detection	Mean Detection	Classification
Recall	Damaged	98.60%	98.37%	100%	99.00%	97.35%	99.6%
	Green	97.20%		100%	94.44%		99.6%
	Red	99.30%		100%	98.61%		99.6%
Precision	Damaged	97.42%	97.68%	100%	86.09%	89.90%	99.6%
	Green	97.20%		100%	85.00%		99.6%
	Red	98.61%		100%	98.61%		99.6%
Accuracy	Damaged	98.60%	98.68%	100%	94.50%	95.34%	99.6%
	Green	98.14%		100%	92.92%		99.6%
	Red	99.29%		100%	98.61%		99.6%
F1-Score	Damaged	97.92%	98.02%	100%	92.09%	93.39%	99.6%
	Green	97.20%		100%	89.47%		99.6%
	Red	98.95%		100%	98.61%		99.6%

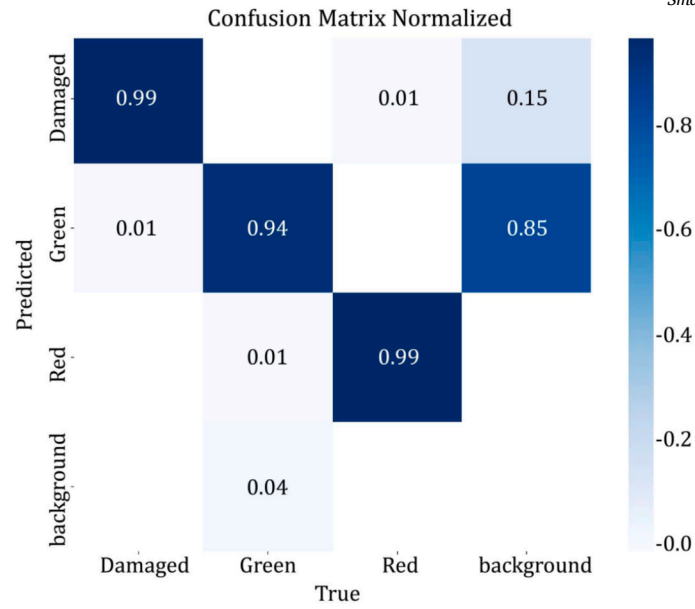


Fig. 8. Normalized confusion matrix of the selected model applied to the testing dataset (216 images). It shows the classification accuracy and misclassification between the three tomato categories.

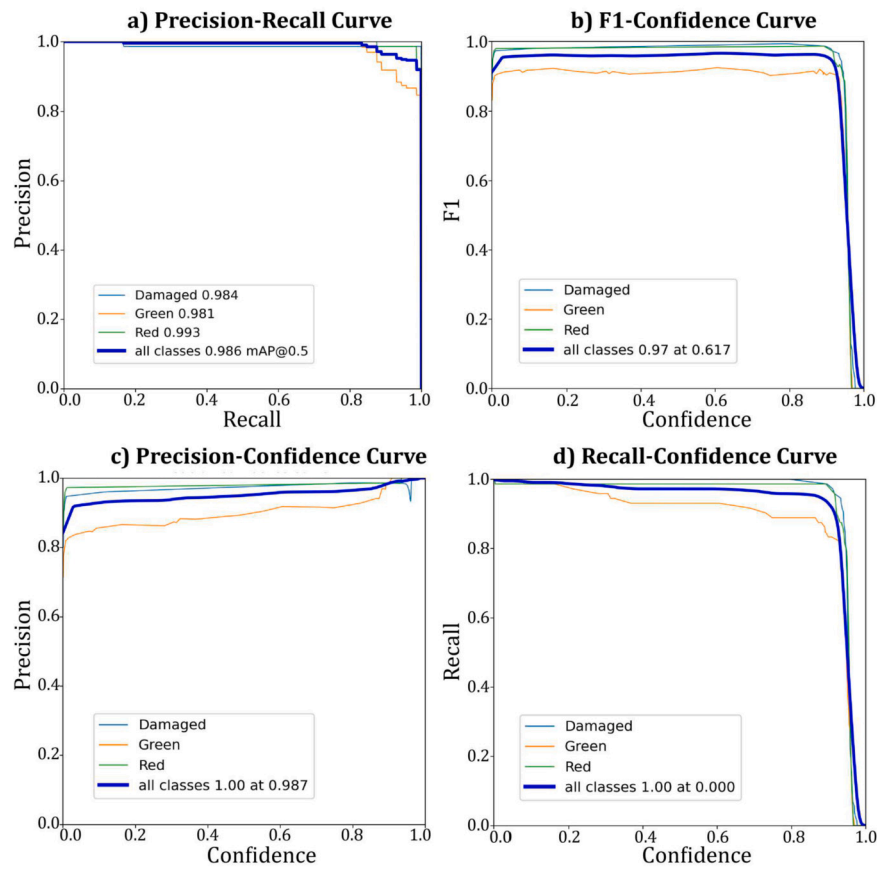


Fig. 9. Performance metrics on the testing dataset. (a) Precision-recall curve; (b) F1-score vs. confidence; (c) Precision vs. confidence; (d) Recall vs. confidence. These graphs demonstrate the model's behaviour across different confidence thresholds.

uct loss. Together, these features not only improve system efficiency but also generate structured data that can support further process optimisation. Examples of these functionalities are illustrated in Fig. 11.

To assess the machine's efficiency, both physically and computationally, tests were conducted using tomatoes from each category. The

results, summarized in Table 5, offer a clear insight into the system's overall performance, taking into account that X corresponds to width and Y to height of the tomato. Table 5 shows the results of the mean squared error (MSE) of 2.7 for X and 3.17 for Y indicating a minimal deviation between the predicted and actual values, suggesting that the

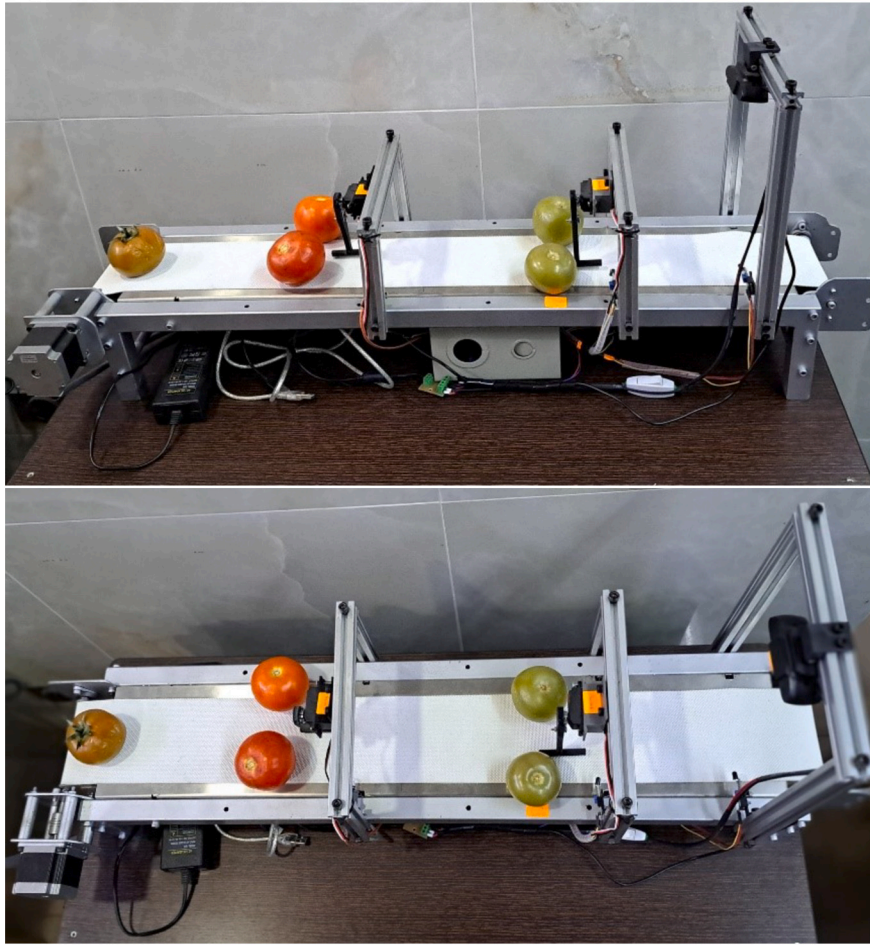


Fig. 10. Real-time classification examples from the system's user interface. (a) Green tomato detection; (b) Red tomato detection; (c) Damaged tomato detection. Includes bounding box and size estimation display.

algorithm is robust and reliable for this application. To evaluate the accuracy of the tomato size estimations, two error metrics were calculated: the relative error and the squared error. The relative error expresses the percentage difference between the estimated and actual measurements, allowing for a normalized interpretation of deviation regardless of tomato size. This makes it especially useful when comparing samples of varying dimensions, as it contextualizes the error in proportion to the true value. The squared error, on the other hand, captures the absolute deviation by squaring the difference between predicted and real dimensions, giving greater weight to larger discrepancies. These metrics underscore the potential of the proposed system as a practical tool for precise, automated measurement in agricultural contexts.

To strengthen the empirical support of the proposed system, a statistical analysis of the size estimation algorithm was conducted. The evaluation included the calculation of the mean relative error, which was found to be 2.89% for the X dimension and 2.83% for the Y dimension, with standard deviations of 1.02% (X) and 1.23% (Y), respectively. These values indicate a tight clustering of estimation errors around their means, suggesting high precision across both dimensions, which is a crucial attribute for automation tasks in post-harvest sorting processes.

Additionally, 95% confidence intervals were computed for both axes. For the X dimension, the interval ranges from 2.41% to 3.37%, while for the Y dimension it spans from 2.25% to 3.41%. These results, shown in Table 6, confirm the reliability and consistency of the proposed algorithm under controlled conditions. The observed minimum errors were 1.70% (X) and 1.07% (Y), while the maximum errors reached 5.54% (X) and 4.89% (Y).

Regarding the colour classification module, no confidence interval was calculated, since the classification accuracy reached 99.6%, with minimal variation across test cases. Misclassification occurred only under extremely low-light conditions (below 5 lux), which are not within the standard operational range of the system. Under appropriate lighting conditions, the classifier maintained robust and consistent performance, as shown in the validation and testing results.

4.1. Lighting conditions experiments

Different experiments were carried out to analyse the performance of the machine under different ranges of illumination and with different sizes of tomatoes, as shown in Table 7, which presents the lux that the machine was subjected to, the class to which the tomato belongs, and the inferences obtained for the width (X) and height (Y), comparing them with the real measurement of each tomato, and calculates the errors in millimetres and percentage. Additionally, Fig. 12 and 13 show the distribution of errors in width and height. For each tomato class, the following analysis was gathered:

- Red class: Relatively consistent and low errors on both axes under good lighting conditions (167-346 lux). At very low lux (4.43), the error tends to increase, particularly on the Y-axis, indicating that this category is sensitive to minimal illumination.
- Green class: Shows stable and accurate behaviour on the X axis, even in low light. However, on the Y axis, the error increases slightly in extreme conditions, which could be related to the colour contrast against the background.

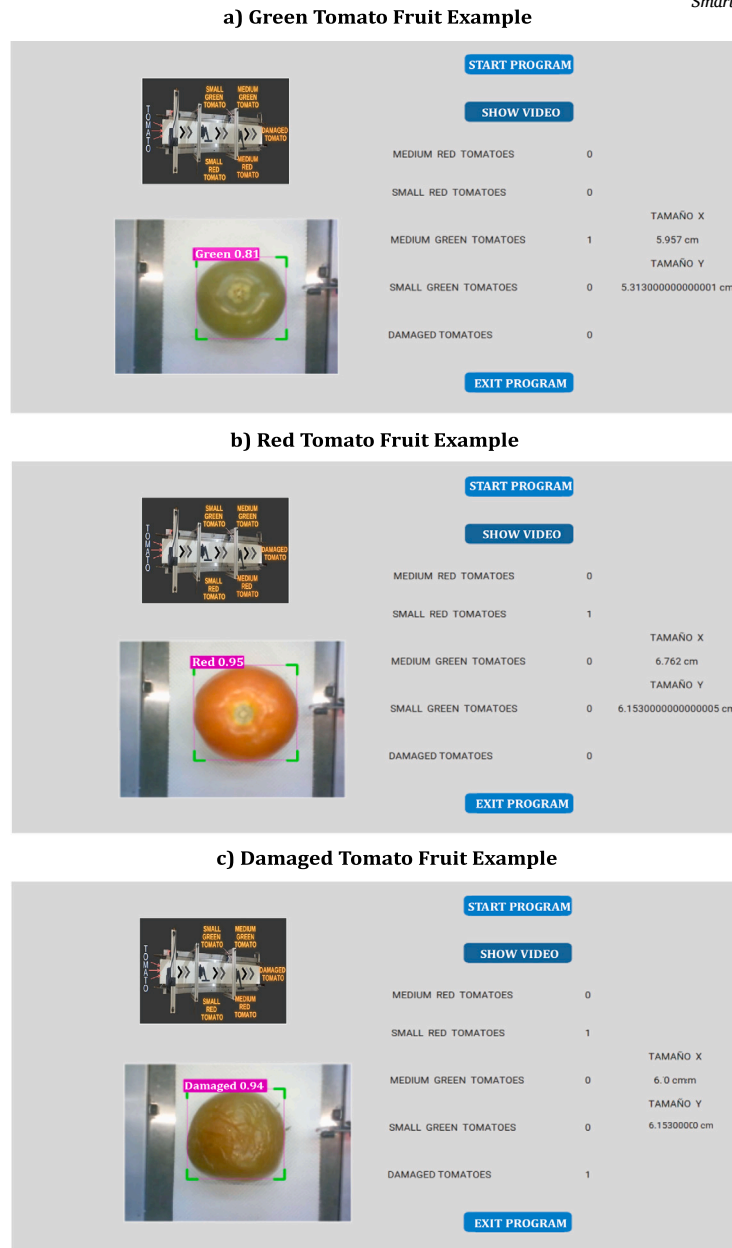


Fig. 11. Experimental results obtained in practice: a) Green Tomato Fruit Example, b) Red Tomato Fruit Example, c) Damaged Tomato Fruit Example.

- **Damaged class:** This category exhibits the highest error and dispersion, particularly on the X-axis at low lux levels. This can be explained by the fact that damage or irregularities on the sample surface can affect the location of the centroid or key edges of the object. In the Y-axis, a wider scatter is also observed, possibly due to physical deformations that interfere with the patterns expected by the model.

The statistical analysis of the X and Y axes, as shown in Figs. 12 and 13, reveals that lighting conditions significantly impact detection accuracy, with 167 lux providing the most stable results across all categories. Under this lighting, both mean error and variance remain low, particularly for green and damaged samples, where X axis errors fall below 1.1 mm and standard deviations are minimal. In contrast, low light conditions (4.43 lux) significantly increase both the mean error and variability, especially for damaged samples, which exhibit the highest X axis error (mean 7.82 mm, maximum 19.05 mm). Y axis errors also exhibit higher dispersion, particularly in red and damaged samples under poor

illumination. Overall, X axis measurements are more consistent, and the system's performance deteriorates notably under insufficient lighting or when dealing with irregular object surfaces, highlighting the need for optimised lighting and possibly enhanced preprocessing for damaged samples.

Furthermore, with a light level of 4.43 lux, there is more dispersion, especially in the damaged category. Considering this, the system is more sensitive to errors due to a lack of contrast or shadows. Within the range of 167-245 lux, it yields better overall results, fewer errors, and no pronounced outliers. This range is the most suitable for data capture. Moreover, finally, at 346.6 lux, the performance remains good; some damaged and red samples begin to exhibit slight deviations, possibly due to saturation or reflections on uneven surfaces.

Regarding colour error produced by lighting conditions, Table 8 shows that the colour classification model performs with perfect accuracy under lighting conditions of 167, 245.2, and 346.6 lux, correctly identifying all red, green, and damaged samples across the three test repetitions. However, at the lowest illumination level (4.43 lux), the

Table 5

Comparison of real tomato diameters measured with a calliper and those estimated by the automated system, including relative and squared errors.

Test	Real Dim. [mm]		Estimated Dim. [mm]		Relative Error [%]		Square Error	
	X	Y	X	Y	X	Y	X	Y
1	42.45	48.25	40.10	47.04	5.54	2.51	5.52	1.46
2	40.85	45.67	41.85	43.48	2.45	4.80	1.00	4.80
3	45.60	52.10	44.37	54.15	2.70	3.93	1.51	4.20
4	47.65	49.60	46.15	51.28	3.15	3.39	2.25	2.82
5	45.20	53.45	43.94	54.32	2.79	1.63	1.59	0.76
6	48.25	55.30	46.87	56.20	2.86	1.63	1.90	0.81
7	43.56	44.15	44.45	44.69	2.04	1.22	0.79	0.29
8	50.50	54.30	52.66	51.88	4.28	4.46	4.67	5.86
9	52.35	55.70	54.10	57.60	3.34	3.41	3.06	3.61
10	54.50	59.85	53.13	62.14	2.51	3.83	1.88	5.24
11	55.60	56.75	54.27	56.14	2.39	1.07	1.77	0.37
12	59.45	60.56	58.35	63.52	1.85	4.89	1.21	8.76
13	60.67	65.60	62.05	62.76	2.27	4.33	1.90	8.07
14	62.75	63.70	60.10	64.88	4.22	1.85	7.02	1.39
15	60.23	63.45	62.95	62.35	4.52	1.73	7.40	1.21
16	60.50	67.12	61.53	68.28	1.70	1.73	1.06	1.35
17	62.85	64.95	61.05	66.39	2.86	2.22	3.24	2.07
18	63.25	66.35	62.15	64.29	1.74	3.10	1.21	4.24
19	65.35	70.75	63.67	68.82	2.57	2.73	2.82	3.72
20	66.72	72.15	68.12	70.59	2.10	2.16	1.96	2.43

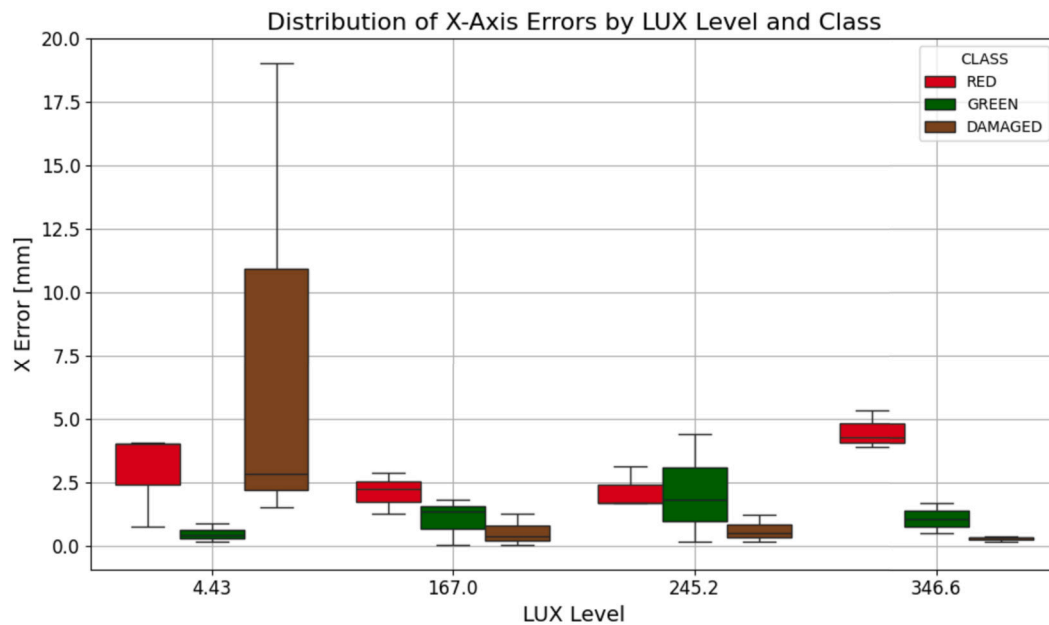


Fig. 12. Distribution of X-axis errors for red, green, and damaged samples under varying Lux levels.

Table 6

Results of confidence interval and standard deviation.

	X %	Y %
Mean Error	2.89	2.83
Confidence Interval (95%)	2.41 – 3.37	2.25 – 3.41
Maximum Error	5.54	4.89
Minimum Error	1.70	1.07
Standard Deviation	1.02	1.23

system misclassifies the damaged class as red in two out of three tests, indicating an apparent sensitivity to poor lighting. These misclassifications highlight that, while the model is robust under medium to high lighting, its performance significantly deteriorates when detecting damaged samples in low-light environments, likely due to reduced visibility of distinguishing features.

To better understand the sequence of the system, the following video shows the machine's operation: <https://youtu.be/9TUhQqAc8M8>.

5. Discussion

This research focused on classifying tomato ripeness and estimating their size under post-harvest conditions. To achieve this goal, a custom dataset comprising 2,145 images was developed, and a classification scheme was defined with three distinct classes: green, red, and damaged. The inclusion of a “damaged” class, not included in other studies, represents a significant advancement, as it addresses visual defects in addition to ripeness, allowing for more selective and robust sorting criteria. This aspect is particularly valuable in post-harvest applications, where both ripeness and external appearance significantly impact the commercial value of the fruit.

Additionally, although the dataset used in this study consisted of 2,145 images, its careful construction and class balance contributed

Table 7

Comparison of software-estimated and real measurements of tomato positions under different lighting levels and categories. Measurement errors in X and Y directions are reported in millimetres and percentages.

LUX	Class	Software [mm]		Real [mm]		X Error		Y Error	
		X	Y	X	Y	[mm]	[%]	[mm]	[%]
TEST 1									
167.00	red	64.05	70.61	66.3	68.4	2.25	3.39	2.21	3.23
	green	69.96	60.27	68.6	63.3	1.36	1.98	3.03	4.79
	damaged	57.96	51.24	58.0	55.5	0.06	0.10	4.36	7.84
346.60	red	70.61	63.21	66.3	68.4	4.31	6.5	5.19	7.59
	green	69.69	60.48	68.6	63.3	1.09	1.59	2.82	4.46
	damaged	57.73	53.27	57.5	65.6	0.17	0.29	4.57	8.22
245.20	red	69.46	65.62	66.3	68.4	3.16	4.77	2.88	4.21
	green	68.77	60.50	68.6	63.3	0.17	0.25	3.24	5.12
	damaged	57.73	50.00	57.9	55.6	0.17	0.29	4.99	8.97
4.43	red	70.38	64.80	66.3	63.3	4.08	6.15	3.72	5.44
	green	68.77	60.06	68.6	63.3	0.17	0.25	3.24	5.12
	damaged	56.35	53.10	57.9	55.6	1.55	2.68	2.5	4.50
TEST 2									
167.00	red	77.97	72.87	76.7	74.9	1.27	1.66	2.03	2.71
	green	65.55	58.8	65.5	63.3	0.05	0.08	5.00	7.89
	damaged	68.70	62.8	67.5	64.5	1.27	1.88	1.70	2.64
346.60	red	78.2	70.56	76.7	74.9	1.5	1.96	4.34	5.79
	green	66.01	59.43	65.5	63.8	0.51	0.78	4.37	6.85
	damaged	67.85	61.74	67.5	64.5	0.35	0.52	2.76	4.28
245.20	red	67.87	63.24	65.2	74.9	1.73	2.26	7.07	9.44
	green	69.93	58.80	65.5	63.8	4.43	6.76	5.00	7.83
	damaged	68.74	61.32	67.5	64.5	1.24	1.84	3.18	4.93
4.43	red	77.5	69.5	76.7	68.4	0.80	1.04	-1.10	-1.61
	green	65.07	57.30	65.5	63.8	0.43	0.66	6.50	10.19
	damaged	64.63	56.50	67.5	64.5	2.87	4.25	8.00	12.40
TEST 3									
167.00	red	68.08	72.87	65.2	67.3	2.88	4.42	5.57	8.28
	green	64.63	56.07	62.8	59.7	1.83	2.91	3.63	6.08
	damaged	67.50	61.32	67.1	65.7	0.40	0.60	4.38	6.67
346.60	red	65.55	57.67	65.2	67.3	0.35	0.54	9.97	14.81
	green	64.53	56.28	62.8	59.7	1.73	2.75	3.42	5.73
	damaged	66.70	61.11	67.1	65.7	0.40	0.60	4.59	6.99
245.20	red	66.93	63.30	65.2	67.3	1.73	2.65	4.00	5.94
	green	64.86	56.28	62.8	59.7	2.06	3.28	3.84	6.43
	damaged	66.7	57.33	67.1	65.7	0.52	0.77	8.37	12.74
4.43	red	69.23	63.63	65.2	67.3	4.03	6.18	3.67	5.45
	green	63.70	60.10	62.8	59.7	0.90	1.43	0.40	0.67
	damaged	48.05	64.05	67.1	65.7	19.05	28.39	1.65	2.51

Table 8

Colour classification inference results and error identification across three test repetitions.

LUX	Class	TEST 1		TEST 2		TEST 3	
		Inference	Result	Inference	Result	Inference	Result
167,00	red	red	no error	red	no error	red	no error
	green	green	no error	green	no error	green	no error
	damaged	damaged	no error	damaged	no error	damaged	no error
346,60	red	red	no error	red	no error	red	no error
	green	green	no error	green	no error	green	no error
	damaged	damaged	no error	damaged	no error	damaged	no error
245,20	red	red	no error	red	no error	red	no error
	green	green	no error	green	no error	green	no error
	damaged	damaged	no error	damaged	no error	damaged	no error
4,43	red	red	no error	red	no error	red	no error
	green	green	no error	green	no error	green	no error
	damaged	red	error	damaged	no error	red	error

significantly to the model's performance. The diversity of acquisition sources and controlled labelling conditions allowed the model to generalise effectively, as evidenced by its consistent accuracy across validation and test sets. While larger datasets can further improve robustness, the results indicate that the selected dataset size was sufficient for the classification and sizing tasks under the tested conditions.

Regarding ripeness classification, several earlier studies, as shown in Table 9, have tackled this task using two or three stages of maturity [24,26,29–31], reporting classification accuracies ranging from 80.6% to 99.1%. While models such as [26] and [29] achieved high accuracy (99.1%), the present study slightly surpasses these results by reaching 99.6% accuracy, while also including the damaged class. This highlights the system's ability to maintain high classification performance

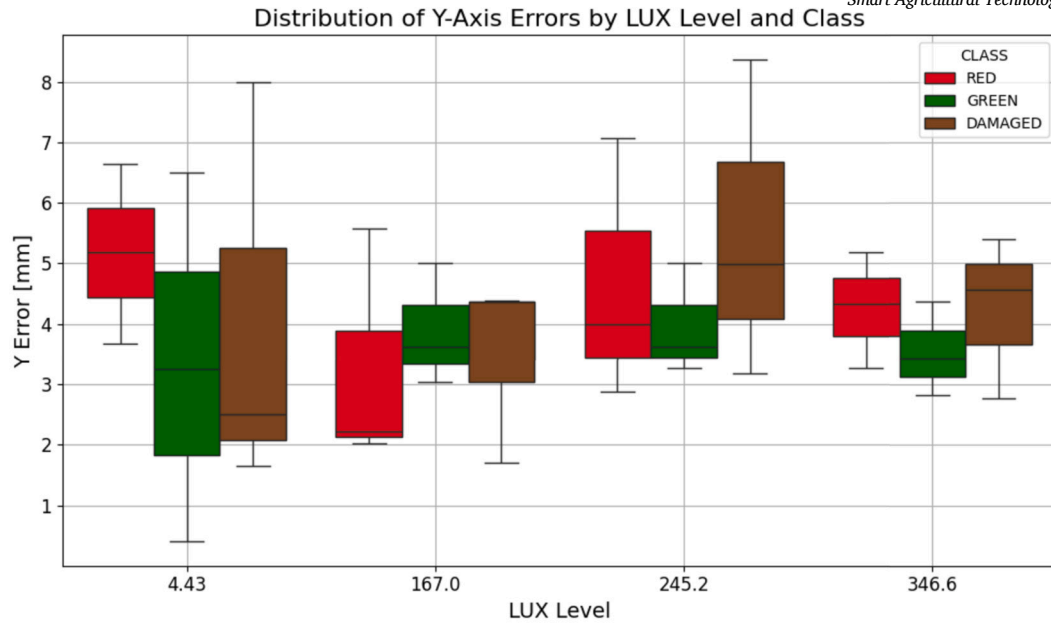


Fig. 13. Distribution of Y-axis errors for red, green, and damaged samples under varying Lux levels.

Table 9

Comparison of Tomato Ripeness and Size Detection Studies using YOLO.

Research	YOLO Version	Ripeness Detection (Classes)	Ripeness Accuracy	Size Classification	Size Accuracy	Implementation / Lighting Control
[29]	YOLOv5 and YOLOv8	Yes (2 stages: ripe and unripe)	99.1%	No	N/A	Yes (Robotic) / No
[35]	YOLOv3	No	N/A	No	N/A	Yes (greenhouse) / Yes
[27]	YOLOv5	Yes (2 stages: TomatoGood and TomatoBad)	90.2%	Yes (size and shape)	N/A	Yes (postharvest) / No
[24]	YOLOv8	Yes (3 stages: green, orange, ripe)	99.1%	No	N/A	No / No
[31]	YOLOv8 + MHSA	Yes (4 classes: Green, Orange, Half Red, Red)	80.6%	No	N/A	No / No
[26]	YOLOv9	Yes (3 stages: Unripe, Semi-ripe, Ripe)	85.6%	No	N/A	No / No
[30]	YOLOv8	Yes (3 stages: Green, Half Ripened, Fully Ripened)	97.23%	No	N/A	Yes (greenhouse video) / No
Current Study	YOLOv8	Yes (3 stages: Green, Ripe, and Damaged)	99.6%	Yes (3 sizes: Small, Medium, Large)	97.1%	Yes (postharvest) / Yes

despite increased class complexity, demonstrating the robustness of the YOLOv8-based architecture and the quality of the dataset.

In terms of size estimation, this work addresses a relatively unexplored dimension. Among the reviewed literature, as shown in Table 9, only [27] touches on size classification, and even then, it does not report a clear accuracy metric. Our system introduces a quantitative classification into three sizes (small, medium, and large) and achieves 97.1% accuracy, thereby offering a dual classification system for ripeness and size. This duality enhances the practical applicability of the model, particularly for automated grading systems where fruit calibration is essential for meeting packaging, pricing, or export quality standards. The integration of both visual maturity and geometric estimation into a single system is a novel contribution with clear industrial implications.

Moreover, physical implementation remains a standard limitation in many existing studies. While some works [27,29,30] report robotic or post-harvest prototypes, they often lack detailed descriptions of the op-

erating environment, particularly regarding lighting conditions, which are known to impact image quality significantly. In contrast, our study presents a fully operational prototype, validated in post-harvest conditions and tested across a defined range of ambient lighting (167–346 lux).

From a performance analysis perspective, additional insight was gained through the error distribution plots as shown in Figs. 12 and 13. It was observed that X axis measurements were more stable, especially under medium to high lighting conditions (167–346 lux), while Y axis measurements exhibited higher variance and outliers. This may be attributed to vertical shadows or the orientation of the fruit, which affects edge detection. This behaviour suggests that further improvements can be made in the calibration of vertical alignment or shadow compensation algorithms, which could strengthen the accuracy of size measurements across both axes.

Beyond the operational performance and technical validation of the system, it is essential to compare this work with related studies, particularly those involving ablation studies. While most recent studies on tomato detection and classification employ ablation experiments to evaluate specific architectural enhancements, the present work adopts a more comprehensive and application-driven approach. For example, Liu et al. [35] introduced YOLO Tomato by incorporating a DenseNet-style dense connection and using Circular Bounding Boxes (C-BBox) instead of rectangular ones, conducting an ablation to assess their individual impact on IoU and precision. Similarly, Zheng et al. [24] enhanced YOLOv8 by adding Large-Kernel Self-Attention (LSKA), Dynamic Upsampling (Dysample), and the Inner-IoU loss function, analysing the contribution of each component. Wang et al. [23] improved YOLOv5n using Coordinate Attention (CA), Weighted IoU loss (WIoU), and optimized anchors via k-means++, performing a classic module-by-module ablation. Yang et al. [25] proposed a lightweight YOLOv8 with Depth-wise Separable Convolutions (DSCov) and a Dual-Path Attention Gate (DPAG), isolating their impact in structured tests.

In contrast, the ablation strategy in this study focuses on training-level variables, like model size (YOLOv8 Nano vs. Small), optimizer type (SGD, Adam, AdamW), batch size (ranging from 8 to 64), and epoch count (25 to 100), and evaluates their influence not only on accuracy and F1-score, but also on physical deployment performance within the sorting machine. Additionally, a novel size estimation algorithm based on real-world calibration was incorporated, enabling two parameter classification (ripeness and diameter in mm) directly from YOLOv8 outputs, something not addressed in the aforementioned works. This holistic and integrative ablation study bridges the gap between algorithmic tuning and system-level optimization, ensuring that model adjustments translate into tangible gains in real-world agricultural settings. As such, the contribution goes beyond improving detection metrics, offering a replicable framework for aligning machine learning configuration with post-harvest operational demands.

In summary, the proposed system demonstrates high precision, practical integration, and robust performance, primarily through its dual classification approach and illumination control. Nonetheless, some limitations remain, including sensitivity to vertical lighting artefacts and the need for controlled positioning to maximise accuracy. Additionally, analysis of the confusion matrix (Fig. 8) and precision-recall metrics revealed that the majority of misclassifications occurred between the red and damaged tomato classes. This confusion arises mainly from surface blemishes and partial spoilage, which under certain lighting conditions can resemble ripeness patterns typical of red tomatoes. For example, although the red class achieved a high precision of 98.61% and recall of 98.61%, some damaged tomatoes were incorrectly labelled as red under low illumination scenarios. In contrast, green tomatoes maintained high recall (94.44%) but lower precision (85.0%), with occasional confusion in transitional ripeness stages or under shadows. These observations highlight the importance of managing lighting variability and improving feature discrimination for damaged categories to guarantee consistent classification under real-world conditions. These aspects will be the focus of future work, including dataset enrichment and the implementation of dynamic illumination compensation techniques, to improve generalisation and adaptability to open-field or greenhouse environments.

6. Conclusions

This study presented an automated system for classifying and sorting kidney tomatoes using YOLOv8 and computer vision. The system achieved a classification accuracy of 99.6% and a size estimation accuracy of 97.1%, demonstrating strong potential for post-harvest applications in industrial environments. The integration of a physical conveyor, proximity sensors, and a servo-driven sorting mechanism reduced manual intervention and enhanced operational efficiency.

A detailed error analysis revealed that most misclassifications occurred between red and damaged tomatoes, primarily due to lighting variability and surface blemishes that confused the model. Green tomatoes were classified with a precision of 85.0% and a recall of 94.44%, while red tomatoes achieved a precision of 98.61% and a recall of 98.61%. Although red tomatoes exhibited strong performance, some were misclassified as damaged when blemishes or lighting artefacts mimicked ripeness indicators. These results highlight the model's overall solid performance, while also underscoring the challenge of accurately distinguishing between red and damaged categories under variable conditions.

Although the proposed prototype was developed for small-scale agricultural settings, its scalability is easy due to its design and the absence of specialised image acquisition hardware. The system utilises commercially available components and a streamlined classification algorithm, enabling replication in larger processing environments. Its inference time supports real-time deployment at higher volumes. In industrial contexts where packaging systems are already in place, the classification phase can be integrated as an independent module, enhancing adaptability without disrupting existing infrastructure, making it a technological solution for both small and industrial-scale tomato producers.

Furthermore, the proposed system is not only scalable but also economically viable for industrial applications. The prototype implemented version had a cost of around USD 500, while the projected cost for an upgraded version ranges from USD 3,500 to 10,000, significantly lower than commercial optical sorters, which often exceed USD 40,000. In addition to this favourable capital investment, the system provides long-term savings by reducing labour costs and product misclassification. Its reliability, maintaining 96.66% classification accuracy under standard conditions, combined with its modular and low-cost design, makes it an attractive solution for small- and medium-scale agricultural producers seeking to modernise their post-harvest operations.

Future work will focus on increasing robustness and generalization. Plans include expanding the dataset to cover partially ripe tomatoes and more diverse lighting scenarios, integrating advanced attention mechanisms (e.g., transformer modules), applying automated hyper-parameter tuning, and comparing YOLOv8 with alternative frameworks such as EfficientDet and RetinaNet. Furthermore, enhancements to the physical sorting system will improve scalability and precision for large-scale agricultural deployment.

CRedit authorship contribution statement

Viviana Moya: Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Michael Guerra:** Validation, Software, Resources, Methodology, Investigation. **Karina Pazmiño:** Visualization, Validation, Supervision, Data curation. **Faruk Abedrabbo:** Writing – original draft, Visualization, Validation, Software, Resources. **Fernando A. Chicaiza:** Writing – review & editing, Supervision, Software, Funding acquisition, Data curation, Conceptualization. **David Pozo-Espín:** Writing – review & editing, Supervision, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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